A

PROJECT REPORT

ON

FLUENCY

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Guided By

Mrs.Dipita Dhande Mam

Submitted By

1.Mrudula Kulkarni

2.Swati Patil

3.Swami Deore

Ashoka Education Foundation's

ASHOKA CENTER FOR BUSINESS AND COMPUTER STUDIES

Certificate

2026-27

This is to certify that --- Mrs.Mrudula Kulkarni, Mrs.Swati Patil, Mr.Swami Deore of Class TYBSc(CS) Semester V Roll No. 23,26,36 - & University Exam Seat No _____ has successfully completed his/her project work entitled Fluency as fulfillment of the requirement of Bachelor of Science (Computer Science) Course, under Savitribai Phule Pune University during the Academic Year 2026-27.

Project Guide

Mrs. Sonali Ingle
Head Of department

Internal Examiner

External Examiner

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INTRODUCTION

FLUENCY is a web-based speech-practice platform designed for people who face difficulties while speaking, such as stammering, hesitation, repetition of words, frequent pauses, slow or uneven speaking pace, and lack of confidence. Some people may feel nervous or uncomfortable while speaking in front of others, which can make regular speaking practice difficult.

FLUENCY provides a supportive and judgement-free environment where users can practise speaking at their own pace. The application provides different practice activities such as Breath & Relax, Reading Practice, Image & Video Practice, Tongue Twisters and AI Conversation. These activities help users practise different aspects of speaking, from controlled reading to spontaneous and real-life conversations.

The system also allows users to record their speech and receive feedback on parameterssuch as speaking speed, pauses, filler words, repetitions and clarity. Users can view theirpractice history, progress graphs and current vs previous performance to understand their progress over time.

The main aim of FLUENCY is to encourage regular speaking practice and help usersgradually improve their speaking fluency, speaking pace and confidence. The system provides constructive feedback to motivate users to continue practising and improving. FLUENCY is a practice and self-improvement platform and is not intended to provide medical diagnosis or replace professional speech therapy.

Preliminary Investigation and its activities

1.Preliminary Investigation and its activities

Preliminary investigation is the initial study carried out to understand the problem, identify the needs of users and determine whether the proposed project is practical and useful. For the FLUENCY project, the investigation focuses on the difficulties faced by people who stammer, hesitate, repeat words or lack confidence while speaking.

1.1 Problem Identification

Many people experience difficulty while speaking. Some people may stammer, hesitate,get stuck on words, repeat sounds or words, take frequent pauses, speak at an uneven pace, or lose confidence while speaking.

Because of these difficulties, they may feel nervous or uncomfortable while speaking in front of other people. Regular speaking practice can help users become more However, practising regularly with another person may not always be convenient. Some comfortable with speaking and become aware of their own speaking patterns.users may also feel uncomfortable or judged while practising in front of others.

Therefore, there is a need for a supportive and private platform where users can practise speaking regularly in a comfortable, non-judgmental environment.

FLUENCY is developed to provide such an environment. It provides different speaking activities, including Reading Practice, Image & Video Practice, AI Conversation, Breath & Relaxation exercises, and Tongue Twister Practice. These activities allow users to practise speaking, observe their speaking patterns, and gradually build confidence.

1.2 Problem Statement / Definition

Many people face difficulties while speaking, such as stammering, hesitation, repetition of sounds or words, frequent pauses, uneven speaking speed, and lack of confidence. These difficulties can make communication uncomfortable, especially in situations such as classrooms, interviews, social conversations, or public speaking.

At present, regular speech practice may require a therapist, teacher, friend, or another person. However, continuous practice with another person may not always be convenient, affordable, or comfortable for everyone. Some individuals may also hesitate to practise because they are afraid of being judged for their speaking

difficulties.Therefore, the problem is to develop a web-based platform that provides a private, supportive, and convenient environment for regular speech practice without judgement.

The proposed system, FLUENCY, addresses this problem by providing different practice activities such as Reading Practice, Image & Video Practice, AI Conversation, Tongue Twister Practice, and Breath & Relaxation. The system also records and analyses the user's speech to provide feedback about aspects such as speaking speed, pauses, repetitions, filler words, and word matching, helping users become more aware of their speaking patterns and practise more effectively.

The main aim is to provide users with an easily accessible platform where they can practise speaking regularly, understand their speaking patterns, and gradually build confidence in communication.

1.3 Purpose / Objective and Goal

The purpose of FLUENCY is to provide a supportive and private web-based platform where users can practise speaking regularly without feeling judged or uncomfortable. The system provides different activities that help users practise different aspects of communication.

FLUENCY also provides speech analysis so that users can become aware of their speaking patterns, such as speaking speed, pauses, repetitions, filler words, and word matching.

The main objectives of the FLUENCY system are:

1. To provide a private and comfortable environment for speech practice.

2. To provide different speaking activities such as Reading Practice, Image & Video Practice, Conversation Practice, and Tongue Twister Practice.

3. To provide Breath & Relaxation exercises before or during practice.

4. To analyse recorded speech and provide useful feedback.

5. To identify speaking patterns such as pauses, repetitions, filler words, speaking speed, and word mismatches.

6. To provide AI-based conversation practice for real-life situations.

7. To maintain the user's practice history and progress.

8. To allow users to select their preferred language for practice and interaction.

9. To help users practise regularly and become more confident while communicating.

Goal

The main goal of FLUENCY is to make regular speaking practice **accessible, convenient, private, and non-judgmental**. The system aims to help users understand their speaking patterns through practice and feedback and gradually improve their confidence in everyday communication.

1.4 Feasibility Study

A feasibility study is conducted to determine whether the proposed FLUENCY system is practical and possible to develop and use. The feasibility of the project can be studied in the following areas.

1. Technical Feasibility

FLUENCY is technically feasible because it can be developed using commonly available web technologies and tools.

The system uses technologies such as:

* HTML, CSS and JavaScript for the frontend
* Python for backend development
* MySQL for database management
* Speech-to-text technology for converting recorded speech into text
* AI-based services for conversation and speech feedback
* Web browsers for accessing the application

These technologies are widely available and can be integrated to develop the required features.

2. Economic Feasibility

The project is economically feasible because it can be developed using commonly available software development tools and technologies. Many of the required development tools are free or open-source.

The system can be developed on an existing computer and accessed through a web browser, reducing the need for special hardware.

Some external AI or speech-processing services may have usage costs depending on the service and level of usage.

3. Operational Feasibility

FLUENCY is operationally feasible because the system is designed to be simple and user-friendly. Users can log in, select a practice activity, record their speech, receive feedback, and view their progress.

The different practice modes allow users to choose an activity according to their needs and practise at their own convenience.

4. Schedule Feasibility

The project is feasible within the available development period because it can be divided into smaller modules such as:

* User authentication
* Profile and settings
* Practice modules
* Speech recording and analysis
* AI conversation
* Progress tracking
* Database integration
* Testing and debugging

These modules can be developed and tested individually and then integrated into the complete system.

Conclusion of Feasibility Study

Based on the technical, economic, operational, and schedule analysis, FLUENCY is considered feasible to develop and implement as a web-based speech practice platform.

1.5 Project Scope and Limitations

Project Scope

The scope of FLUENCY includes providing users with a platform for regular speech practice and confidence building.

The system includes the following major functionalities:

1. User Registration and Login

Users can create an account and securely access their profile.

2. User Profile and Settings

Users can manage their personal details, avatar, age group, preferred language, and other preferences.

3. Breath & Relax

Users can perform timed breathing and relaxation exercises before practising.

4. Reading Practice

Users can read paragraphs of different lengths and difficulty levels while recording their speech.

5.Image & Video Practice

Users can observe existing images or videos and describe what they see.

6. Tongue Twister Practice

Users can practise different tongue twisters and receive speech-related feedback.

7. AI Conversation

Users can practise conversations based on different real-life scenarios with an AI conversation partner.

8. Speech Analysis

The system analyses recorded speech and provides information such as speaking speed, pauses, repetitions, filler words, and word matching.

9. My Progress

Users can view their practice history and progress-related information.

10. Videos on Therapy

Users can watch educational and awareness-related videos related to speech and confidence.

11. Multilingual Support

Users can select a supported language for the interface and speech practice.

12. Admin/User Management

The administrator can view registered users and their overall practice information.

Limitations

Although FLUENCY provides several speech-practice features, it has some limitations:

1. The accuracy of speech analysis depends on the quality of speech recognition and microphone input.

2. Background noise or poor microphone quality may affect the accuracy of the recorded transcript and analysis.

3. AI-generated feedback may not always perfectly understand every speech situation.

4. The system is a practice and confidence-building platform, not a replacement for a professional speech therapist or medical treatment.

5. Speech recognition and AI features may depend on internet connectivity and external services.

6. The accuracy of language recognition may vary between different supported languages.

7. The system currently focuses on speech practice and awareness rather than providing clinical diagnosis or treatment.

8. The quality of progress analysis depends on the amount and quality of practice data recorded by the user.

Overall Scope

Thus, FLUENCY focuses on providing a convenient, private, and supportive environment for regular speaking practice, while professional diagnosis or treatment remains outside the scope of the system.

2. Requirement Specification

2.1 System Requirements

The following hardware and software requirements are required to develop and run the FLUENCY web application.

Hardware Requirements

Server / Device Requirements

FLUENCY can be run on a local development server using a computer or laptop

Component	Minimum Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB minimum, 8 GB recommended
Storage	At least 10 GB free space
Microphone	Required for speech recording and speech analysis
Camera	Optimal,required only for camera-based feature if enabled
Internet	Required for AI and speech-processing feature

Backend Server: Local server using Python/Flask

Database Server: MySQL 8.0

Host: Localhost / `127.0.0.1`

Database: `fluency_db`

Port: MySQL uses port `3306`; the Flask application runs on its configured local web port.

Client Device: Laptop, desktop, or other device with a modern web browser.

Browser: Google Chrome, Microsoft Edge, Firefox, or another modern browser supporting microphone access.

Software Requirements

Software	Requirement
Operating System	Windows 10/11 or equivalent
Frontend	HTML5,CSS3,JavaScript
Backend	Python with flask
Databse	MySQL 8.0
Database Driver	PyMySQL
Browser	Modern web browser
Code Editor	Visual Studio Code/similar editor
Database Tool Workbench	MySQL Workbench
Version Control	Git/GitHub,if required
Internet	Required for external AI/Speech services

2.2 Technical Requirements

Programming Languages

The following programming languages and technologies are used in FLUENCY:

1. HTML5 – Used to create the structure of web pages.

2. CSS3 – Used for styling, layout, colors, responsiveness, and visual design.

3. JavaScript – Used for client-side interaction, recording controls, dynamic content, and communication with the backend.

4. Python – Used for backend development and application logic.

5. SQL – Used for storing, retrieving, and managing data in MySQL.

Frameworks and Tools

Flask: Python web framework used for backend development.

MySQL: Relational database management system used to store user and practice data.

MySQL Workbench: Used for database management and SQL queries.

PyMySQL: Used to connect the Python application with MySQL.

Speech-to-Text: Used to convert recorded user speech into text for analysis.

AI Services: Used for AI conversation and speech-related feedback.

Web Browser APIs: Used for microphone/audio recording and speech-related functionality.

Visual Studio Code:Used as the development environment.

2.3 Functional Requirements

Functional requirements describe what the FLUENCY system should do.

1. User Registration and Login

* The system shall allow new users to create an account.

* The system shall allow registered users to log in.

* The system shall authenticate users before providing access to their personal data.

* The system shall allow users to log out.

2. User Profile Management

* The system shall allow users to view and update their profile.

* Users shall be able to select an avatar.

* Users shall be able to select their age group.

* Users shall be able to select their preferred language.

* The system shall store updated profile information in the database.

3. Breath & Relaxation

* The system shall provide a breathing/relaxation activity.

* Users shall be able to select a timer duration.

* The system shall run the selected timer.

4. Reading Practice

* The system shall provide reading paragraphs.

* Users shall be able to select difficulty and paragraph length.
* The system shall display content according to the selected options and user age group.
* Users shall be able to record their speech.
* The system shall convert the recording into text.
* The system shall compare the spoken text with the displayed paragraph.
* The system shall identify missing, extra, repeated, or mismatched words.
* The system shall provide speech-related feedback.

5. Image & Video Practice

* The system shall provide image and video-based speaking activities.
* Users shall select a difficulty level.
* Users shall observe the displayed visual and describe it.
* The system shall provide relevant prompts and guiding questions.
* Users shall be able to record their response.
* The system shall analyse the recorded speech.

6. Tongue Twister Practice

* The system shall provide different tongue twisters.
* Users shall be able to select/change the tongue twister.
* Users shall record themselves reading the tongue twister.
* The system shall compare the spoken words with the expected tongue twister.
* The system shall provide relevant speech feedback.

7. AI Conversation

* The system shall provide different real-life conversation scenarios.
* Users shall be able to select a scenario.
* The AI shall respond according to the selected scenario.
* The AI shall understand the user's response and continue the conversation.
* The system shall analyse the user's speech during the conversation where supported.
* The conversation shall continue until the user chooses to end it.

8. Speech Analysis

The system shall analyse recorded speech and, where reliably available, provide:

* Speaking duration

* Number of words spoken

* Speaking speed

* Long pauses

* Repetitions

* Filler words

* Word matching

* Other relevant speech observations

The system shall provide supportive and specific feedback based on the detected speech patterns.

9. My Progress

* The system shall maintain practice history.

* Users shall be able to view their previous practice sessions.

* Users shall be able to view progress-related information.

* Saved recordings shall be available for replay when available.

* Progress information shall be associated with the correct user account.

10. Videos on Therapy

* The system shall provide educational and awareness videos.

* Users shall be able to watch videos related to speaking, confidence, and communication.

* The module is intended for learning and awareness, not speech diagnosis.

11. Multilingual Support

* The system shall allow users to select a supported language.

* User-facing content shall follow the selected language.

* Practice content shall use the selected language.

* Speech recognition shall use the selected language where supported.

* AI conversation and feedback shall follow the selected language.

12. Admin Functions

* The system shall provide an admin login.

* The administrator shall be able to view registered users.

* The administrator shall be able to view overall user practice information.

* User data shall remain separated according to user accounts.

2.4 Data Requirements

The FLUENCY system requires data to manage users, practice sessions, conversations, progress, and bug reports.

The system stores the following data:

* User Data: Name, email, password, age/DOB, phone, language and avatar.

* Practice Data: Practice mode, difficulty, duration, prompt, transcript and date.

* Recording Data: Audio recording, type and duration.

* Speech Analysis Data: Speaking speed, fillers, pauses, repetitions, clarity, consistency and feedback.

* Tongue Twister Data: Tongue-twister text, category and difficulty.

* Conversation Data: Scenario, AI persona, dialogue and speech metrics.

* Progress Data: Streak, total sessions, practice time and performance.

* Preference Data: Mascot, AI partner, speaking goals and analysis settings.

* Bug/Feedback Data: Issue, description, severity, page and status.

3. Database Design

3.1 Identify End Users of the System

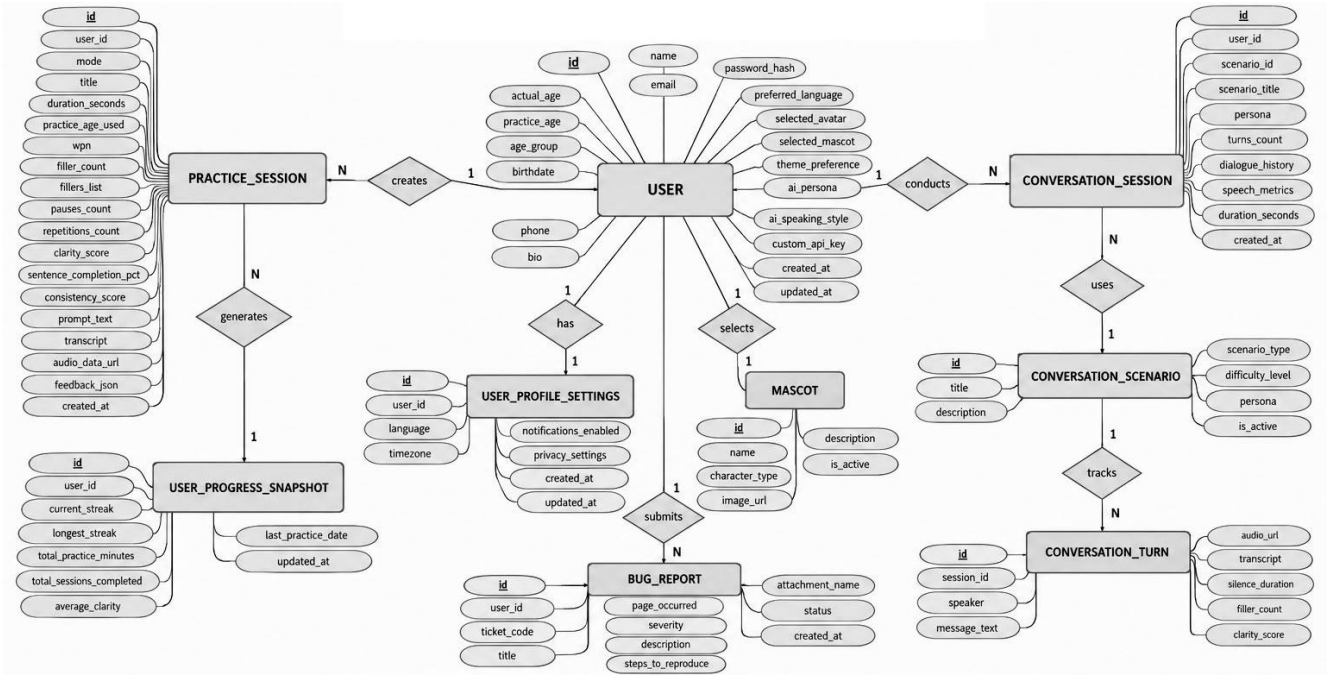
The main end user of FLUENCY is:

User

A person who wants to practise speaking, improve fluency and speaking pace, and build confidence.

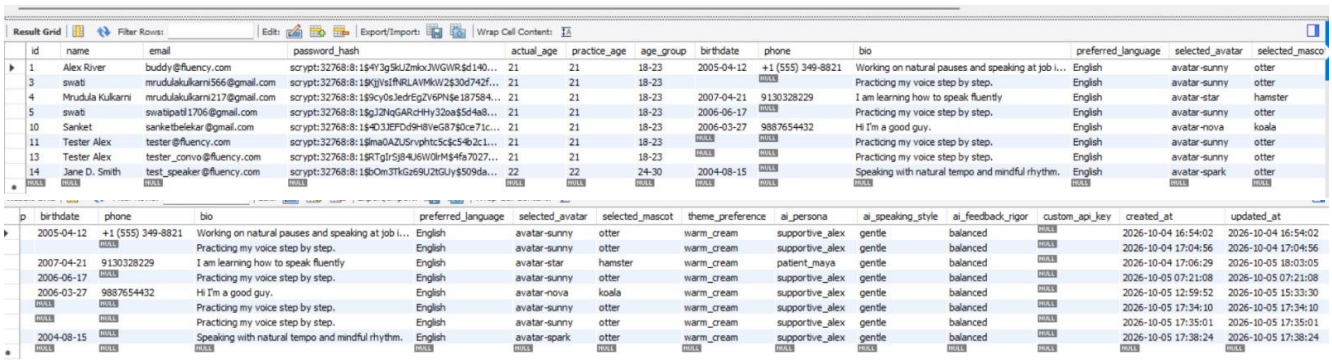
The Administrator/Developer may manage system content, bugs and application-related data if an admin module is implemented.

ER Diagram:-



1. Users

Stores user account and profile information.



id	name	email	password_hash	actual_age	practice_age	age_group	birthdate	phone	bio	preferred_language	selected_avatar	selected_masco
1	Alex River	buddy@fluency.com	scrypt:32768:8:1$4Y3gSkUZmkxJWGWR$d140…	21	21	18-23	2005-04-12	+1 (555) 349-8821	Working on natural pauses and speaking at job i…	English	avatar-sunny	otter
3	swati	mrudulakulkarni566@gmail.com	scrypt:32768:8:1$KjjVs1fNRLAVMkVf2$30d742f…	21	21	18-23	NULL	NULL	Practicing my voice step by step.	English	avatar-sunny	otter
4	Mrudula Kulkarni	mrudulakulkarni217@gmail.com	scrypt:32768:8:1$9cy0sJedrEgZV6PN$e187584…	21	21	18-23	2007-04-21	9130328229	I am learning how to speak fluently	English	avatar-star	hamster
5	swati	swatipatil1706@gmail.com	scrypt:32768:8:1$gJ2fNqGARcHHy32oa$5d4a8…	21	21	18-23	2006-06-17	NULL	Practicing my voice step by step.	English	avatar-sunny	otter
10	Sanket	sanketbelekar@gmail.com	scrypt:32768:8:1$4D3JEFDd9H8VeG87$0ce71c…	21	21	18-23	2006-03-27	9887654432	Hi I'm a good guy.	English	avatar-nova	koala
11	Tester Alex	tester@fluency.com	scrypt:32768:8:1$ImaOAZUSrvphtcSc$c54b2c1…	21	21	18-23	NULL	NULL	Practicing my voice step by step.	English	avatar-sunny	otter
13	Tester Alex	tester_convo@fluency.com	scrypt:32768:8:1$RTgJrSj84U6W0lrM$4fa7027…	21	21	18-23	NULL	NULL	Practicing my voice step by step.	English	avatar-sunny	otter
14	Jane D. Smith	test_speaker@fluency.com	scrypt:32768:8:1$bOm3TkGz69U2tGUy$509da…	22	22	24-30	2004-08-15	NULL	Speaking with natural tempo and mindful rhythm.	English	avatar-spark	otter
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

p	birthdate	phone	bio	preferred_language	selected_avatar	selected_mascot	theme_preference	ai_persona	ai_speaking_style	ai_feedback_rigor	custom_api_key	created_at	updated_at
	2005-04-12	+1 (555) 349-8821	Working on natural pauses and speaking at job i…	English	avatar-sunny	otter	warm_cream	supportive_alex	gentle	balanced	NULL	2026-10-04 16:54:02	2026-10-04 16:54:02
	NULL	NULL	Practicing my voice step by step.	English	avatar-sunny	otter	warm_cream	supportive_alex	gentle	balanced	NULL	2026-10-04 17:04:56	2026-10-04 17:04:56
	2007-04-21	9130328229	I am learning how to speak fluently	English	avatar-star	hamster	warm_cream	patient_maya	gentle	balanced	NULL	2026-10-04 17:06:29	2026-10-05 18:03:05
	2006-06-17	NULL	Practicing my voice step by step.	English	avatar-sunny	otter	warm_cream	supportive_alex	gentle	balanced	NULL	2026-10-05 07:21:08	2026-10-05 07:21:08
	2006-03-27	9887654432	Hi I'm a good guy.	English	avatar-nova	koala	warm_cream	supportive_alex	gentle	balanced	NULL	2026-10-05 12:59:52	2026-10-05 15:33:30
	NULL	NULL	Practicing my voice step by step.	English	avatar-sunny	otter	warm_cream	supportive_alex	gentle	balanced	NULL	2026-10-05 17:34:10	2026-10-05 17:34:10
	NULL	NULL	Practicing my voice step by step.	English	avatar-sunny	otter	warm_cream	supportive_alex	gentle	balanced	NULL	2026-10-05 17:35:01	2026-10-05 17:35:01
	2004-08-15	NULL	Speaking with natural tempo and mindful rhythm.	English	avatar-spark	otter	warm_cream	supportive_alex	gentle	balanced	NULL	2026-10-05 17:38:24	2026-10-05 17:38:24
	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

id	name	email	actual_age	practice_age	age_group	preferred_language	created_at
1	Alex River	buddy@fluency.com	21	21	18-23	English	2026-10-04 16:54:02
3	swati	mrudulakulkarni566@gmail.com	21	21	18-23	English	2026-10-04 17:04:56
4	Mrudula Kulkarni	mrudulakulkarni217@gmail.com	21	21	18-23	English	2026-10-04 17:06:29
5	swati	swatiipatil1706@gmail.com	21	21	18-23	English	2026-10-05 07:21:08
10	Sanket	sanketbelekar@gmail.com	21	21	18-23	English	2026-10-05 12:59:52
11	Tester Alex	tester@fluency.com	21	21	18-23	English	2026-10-05 17:34:10
13	Tester Alex	tester_convo@fluency.com	21	21	18-23	English	2026-10-05 17:35:01
14	Jane D. Smith	test_speaker@fluency.com	22	22	24-30	English	2026-10-05 17:38:24
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
name	varchar(120)	NO		NULL	
email	varchar(120)	NO	UNI	NULL	
password_hash	varchar(256)	NO		NULL	
actual_age	int	YES		NULL	
practice_age	int	YES		NULL	
age_group	varchar(20)	YES		NULL	
birthdate	varchar(20)	YES		NULL	
phone	varchar(30)	YES		NULL	
bio	text	YES		NULL	
preferred_lang...	varchar(30)	YES		NULL	
selected_avatar	varchar(50)	YES		NULL	
selected_mascot	varchar(30)	YES		NULL	
theme_prefere...	varchar(30)	YES		NULL	
ai_persona	varchar(40)	YES		NULL	
ai_speaking_st...	varchar(30)	YES		NULL	
ai_feedback_ri...	varchar(30)	YES		NULL	
custom_api_key	varchar(256)	YES		NULL	
created_at	datetime	YES		NULL	
updated_at	datetime	YES		NULL	

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
user_id	int	NO	MUL	NULL	
mode	varchar(30)	NO	MUL	NULL	
title	varchar(200)	YES		NULL	
duration_seconds	int	YES		NULL	
practice_age_used	int	YES		NULL	
wpm	float	YES		NULL	
filler_count	int	YES		NULL	
fillers_list	json	YES		NULL	
pauses_count	int	YES		NULL	
repetitions_count	int	YES		NULL	
clarity_score	float	YES		NULL	
sentence_complet...	float	YES		NULL	
consistency_score	float	YES		NULL	
prompt_text	text	YES		NULL	
transcript	text	YES		NULL	
audio_data_url	mediumtext	YES		NULL	
feedback_json	json	YES		NULL	
created_at	datetime	YES	MUL	NULL	

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
user_id	int	NO	UNI	NULL	
current_streak	int	YES		NULL	
longest_streak	int	YES		NULL	
total_practice_minutes	float	YES		NULL	
total_sessions_completed	int	YES		NULL	
average_clarity	float	YES		NULL	
last_practice_date	varchar(20)	YES		NULL	
updated_at	datetime	YES		NULL	

2. Practice Sessions

Stores information about individual practice sessions.

Important fields:

* id

* user_id

* mode
* title
* duration_seconds
* practice_age_used
* wpm
* filler_count
* fillers_list
* pauses_count
* repetitions_count
* clarity_score
* sentence_completion_pct
* consistency_score
* prompt_text
* transcript
* audio_data_url
* feedback_json
* created_at

3. Conversation Sessions

Stores information about AI conversation sessions.

Important fields:

* id
* user_id
* scenario_id
* scenario_title
* persona
* turns_count
* dialogue_history
* speech_metrics

* duration_seconds

* created_at

4. User Progress Snapshots

Stores the user's overall practice progress.

Important fields:

* id

* user_id

* current_streak

* longest_streak

* total_practice_minutes

* total_sessions_completed

* average_clarity

* last_practice_date

* updated_at

5. Bug Reports

Stores issues reported by users.

Important fields:

* id

* user_id

* ticket_code

* title

* page_occurred

* severity

* description

* steps_to_reproduce

* attachment_name

* status

* created_at

3.3 Tables,Fields and Relationships

1.User

Field	Description
id	Unique user ID PK
name	User’s name
email	User’s email Unique
password_hash	Securely stored password hash
actual_age	User’s actual age
practice_age	Age used for practice personalization
age_group	User’s age group
birthday	Date of birth
phone	Phone number
bio	User biography/goals
preferred_language	Selected language
selected_avatar	Selected avatar
selected_mascot	Selected mascot
theme_preference	Selected theme
Ai_person	AI person preference
ai_speaking_style	AI speaking style
ai_feedback_rigor	AI feedback preference
Custom_api_key	Optimal custom API key
created_at	Account creation time

Parent Table	Child Table	RelationShip

users	practice_sessions	1:N
users	conversation_sessions	1:N
users	user_progress	1:1
users	bug_reports	1:N

3.4 Normalize Database

Database normalization is the process of organizing data to reduce *data redundancy, duplication, and inconsistency and to improve data integrity.

The FLUENCY database is organized so that different types of information are stored in their appropriate tables.

First Normal Form (1NF)

A table is in 1NF when:

* Each column contains atomic values.

* Each record is uniquely identifiable.

* There are no repeating groups.

In FLUENCY, each table has a unique primary key such as id. User details, practice information, conversation information, progress information, and bug reports are stored as separate records.

Therefore, the database follows the principles of *1NF*.

Second Normal Form (2NF)

A table is in 2NF when:

* It is already in 1NF.

* Non-key attributes depend on the complete primary key.

In FLUENCY, the tables use primary keys such as id, and the attributes of each table describe the entity represented by that table.

For example, user profile information is stored in users, while practice-specific information is stored in practice_sessions.

Therefore, the database follows the principles of *2NF*.

Third Normal Form (3NF)

A table is in 3NF when:

* It is already in 2NF.

* Non-key attributes do not depend on other non-key attributes.

FLUENCY separates different types of information into appropriate tables:

* User information → users

* Practice information → practice_sessions

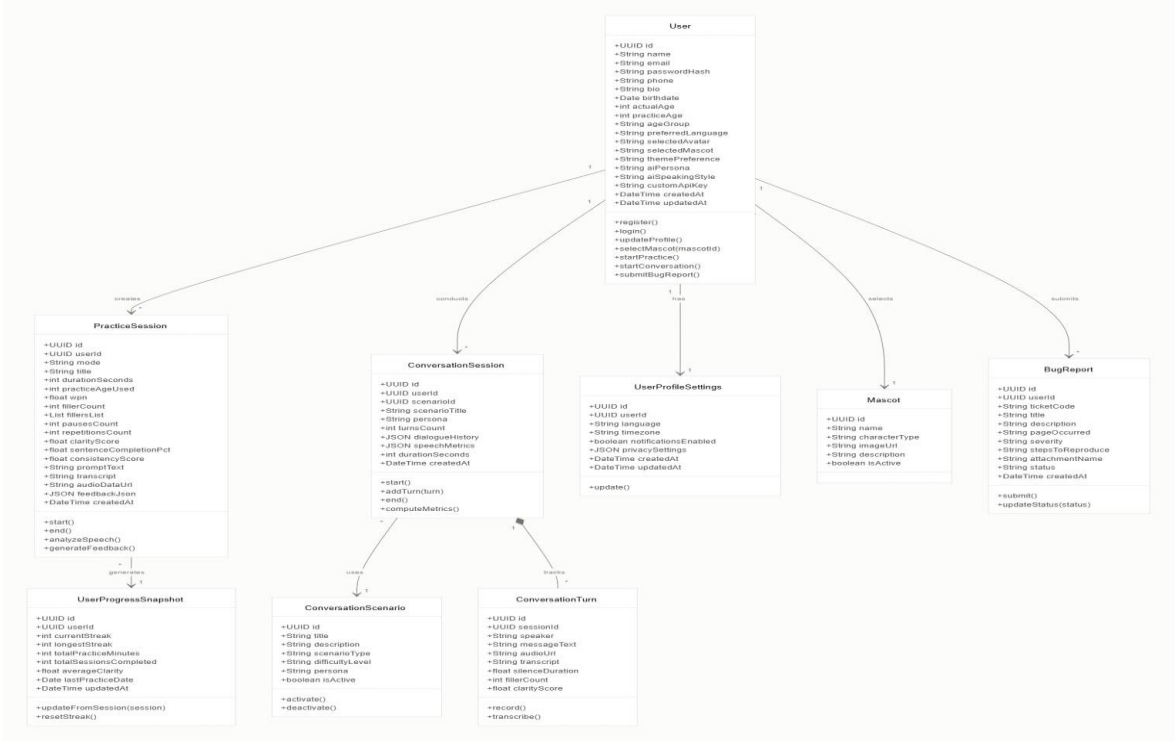
* Conversation information → conversation_sessions

* Progress information → user_progress_snapshots

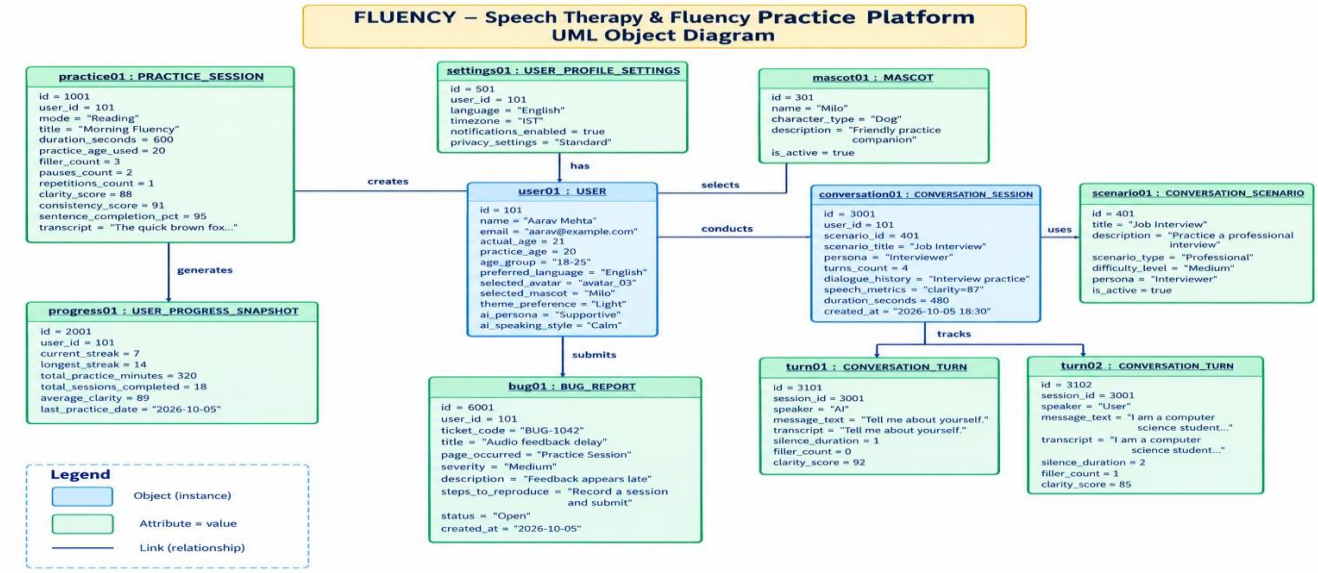
* Bug information → bug_reports

4.System Desing

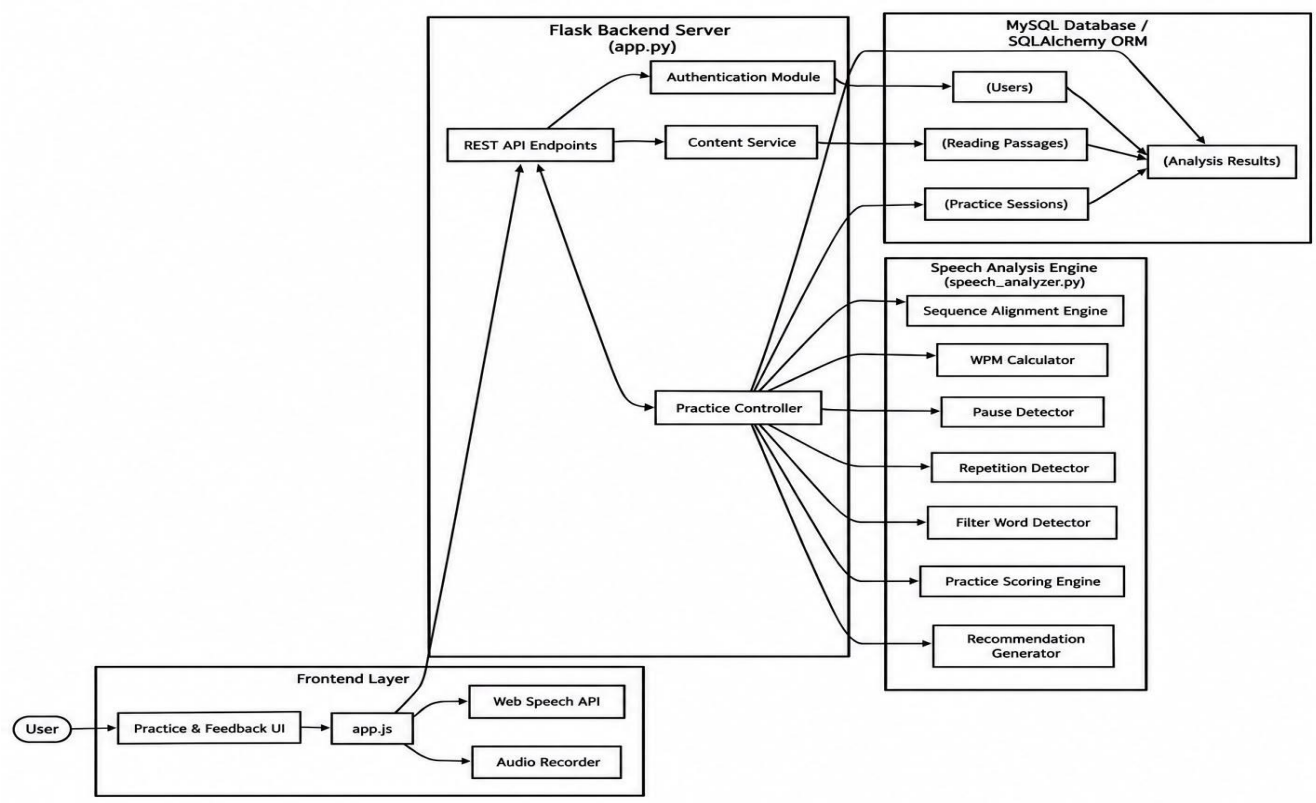
4.1 Class Diagram



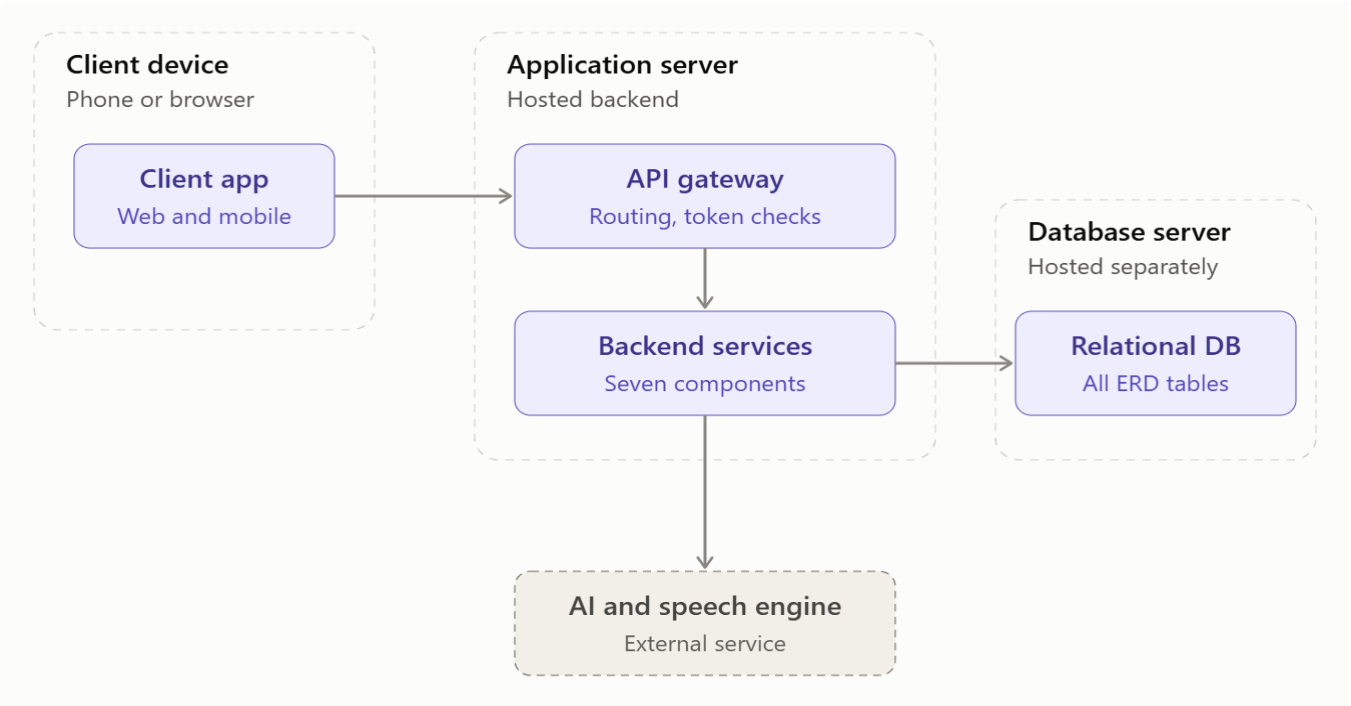
4.2 Object Diagram



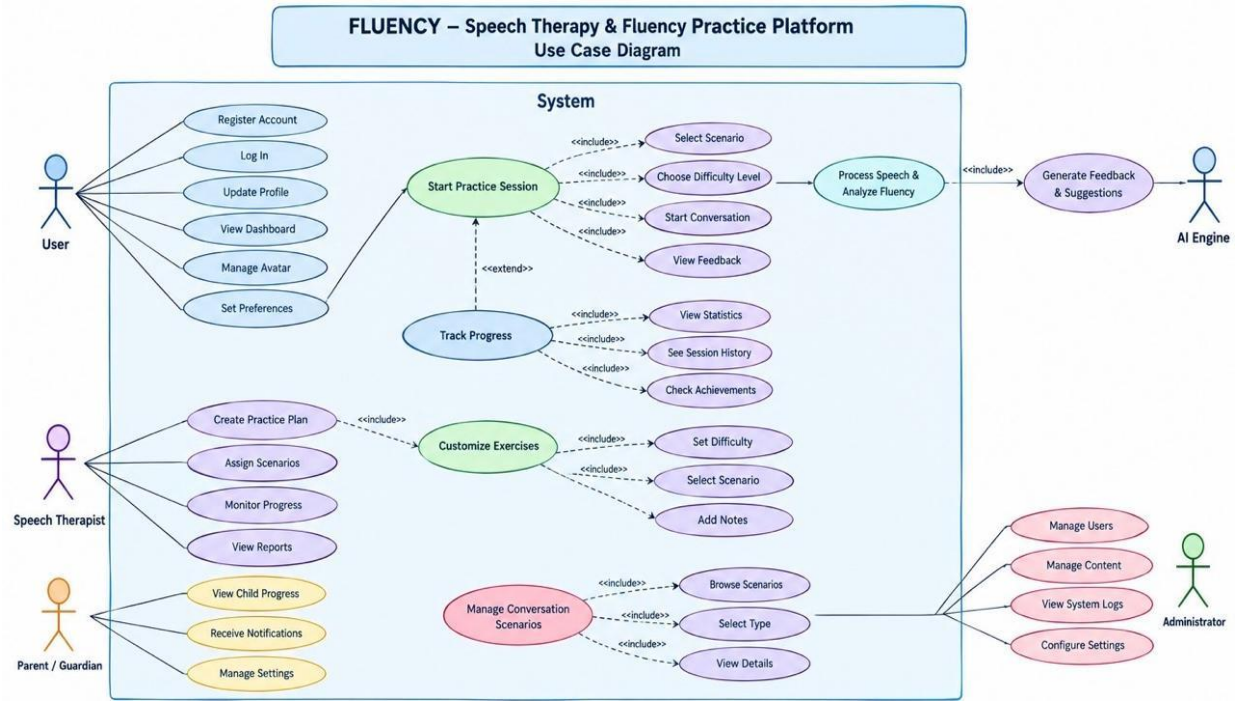
4.3 Component Diagram



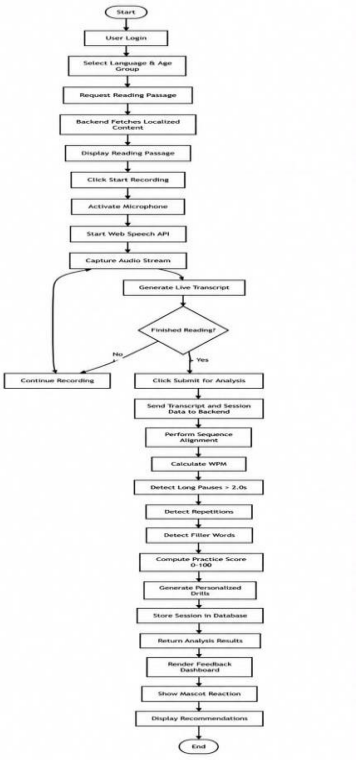
4.4 Deployment Diagram



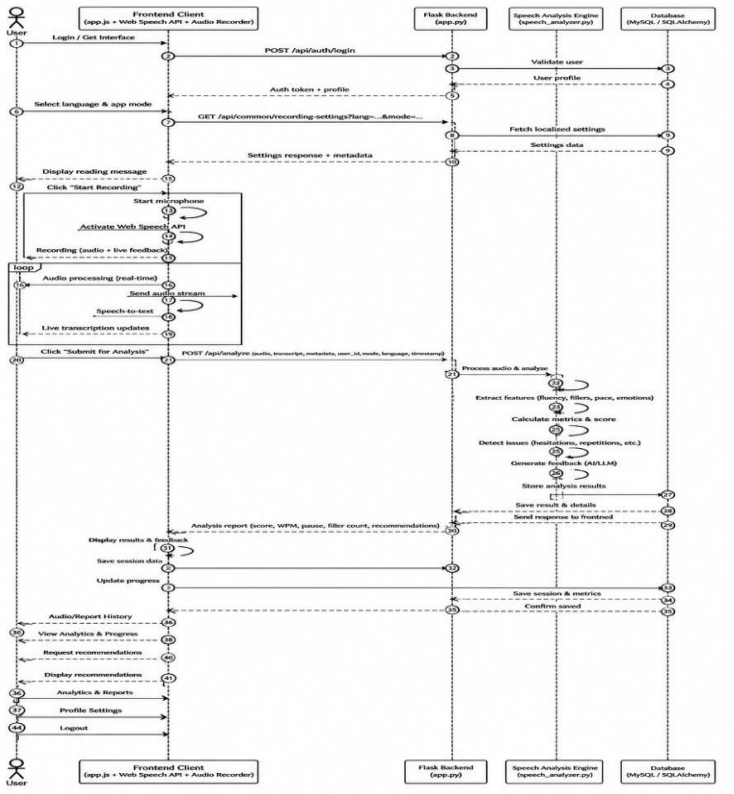
4.5 Use Case Diagram



4.6 Activity Diagram

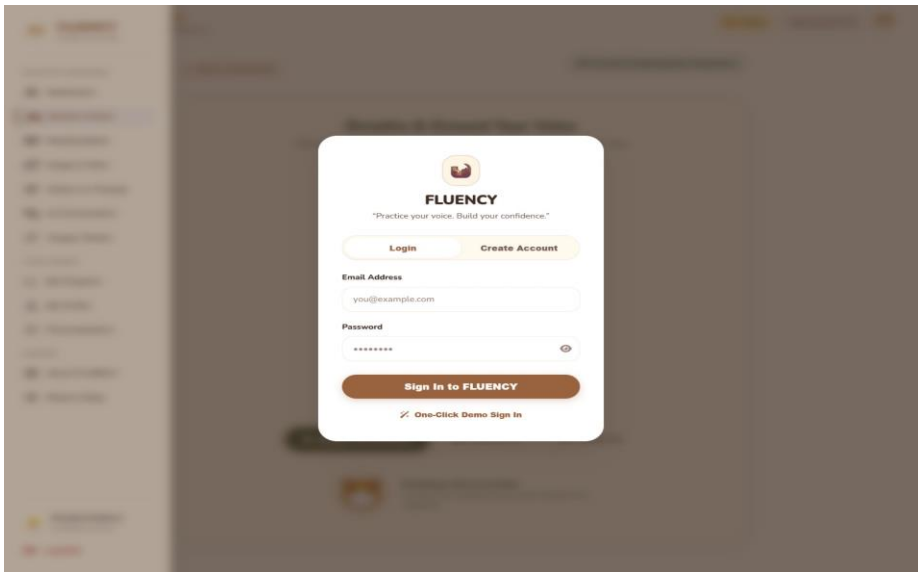


4.7 Sequence Diagram

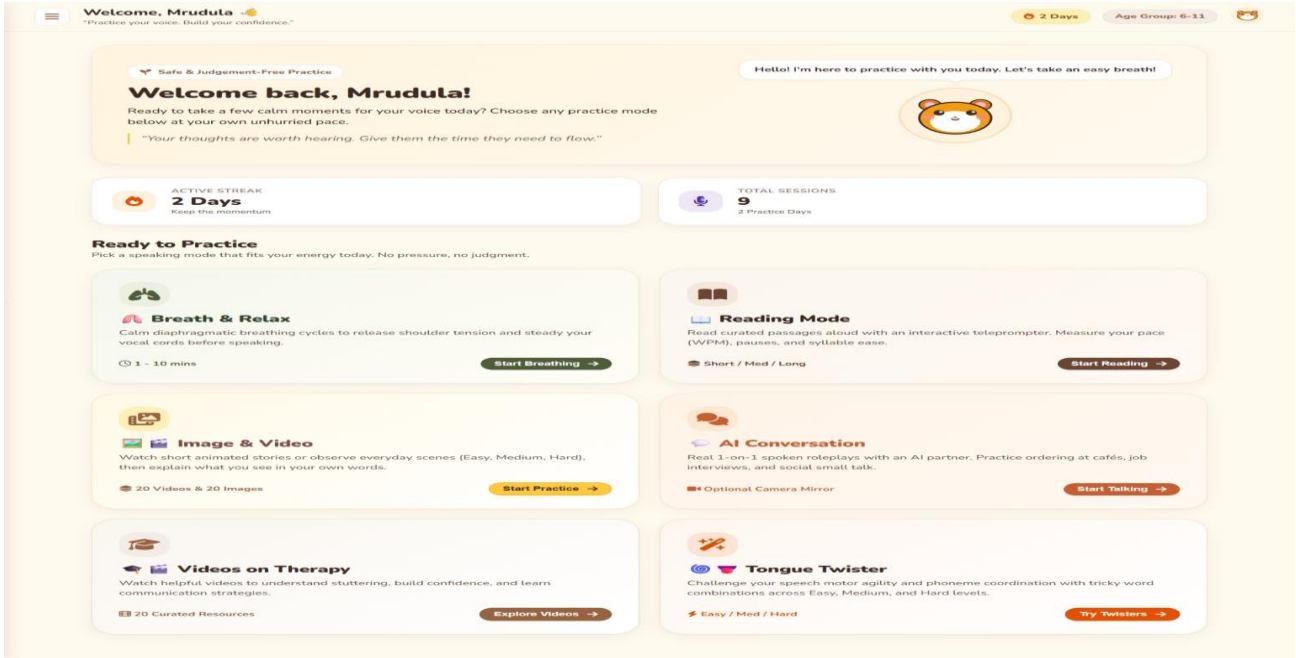

User
Frontend Client
(app.js + Web Speech API + Audio Recorder)
Flask Backend
(app.py)
Speech Analysis Engine
(speech_analyzer.py)
Database
(MySQL / SQLAlchemy)
Login / Get Interface
POST /api/auth/login
Validate user
User profile
Auth token + profile
Select language & app mode
GET /api/common/recording-settings?lang=...&mode=...
Fetch localized settings
Settings data
Settings response + metadata
Display reading message
Click "Start Recording"
Start microphone
Activate Web Speech API
Recording (audio + live feedback)
loop
Audio processing (real-time)
Send audio stream
Speech-to-text
Live transcription updates
Click "Submit for Analysis"
POST /api/analyze (audio, transcript, metadata, user_id, mode, language, timestamp)
Process audio & analyze
Extract features (fluency, fillers, pace, emotions)
Calculate metrics & score
Detect issues (hesitations, repetitions, etc.)
Generate feedback (AI/LLM)
Store analysis results
Save result & details
Send response to frontend
Analysis report (score, WPM, pause, filler count, recommendations)
Display results & feedback
Save session data
Update progress
Save session & metrics
Confirm saved
Audio/Report History
View Analytics & Progress
Request recommendations
Display recommendations
Analytics & Reports
Profile Settings
Logout

5.Input and Output screen Layout

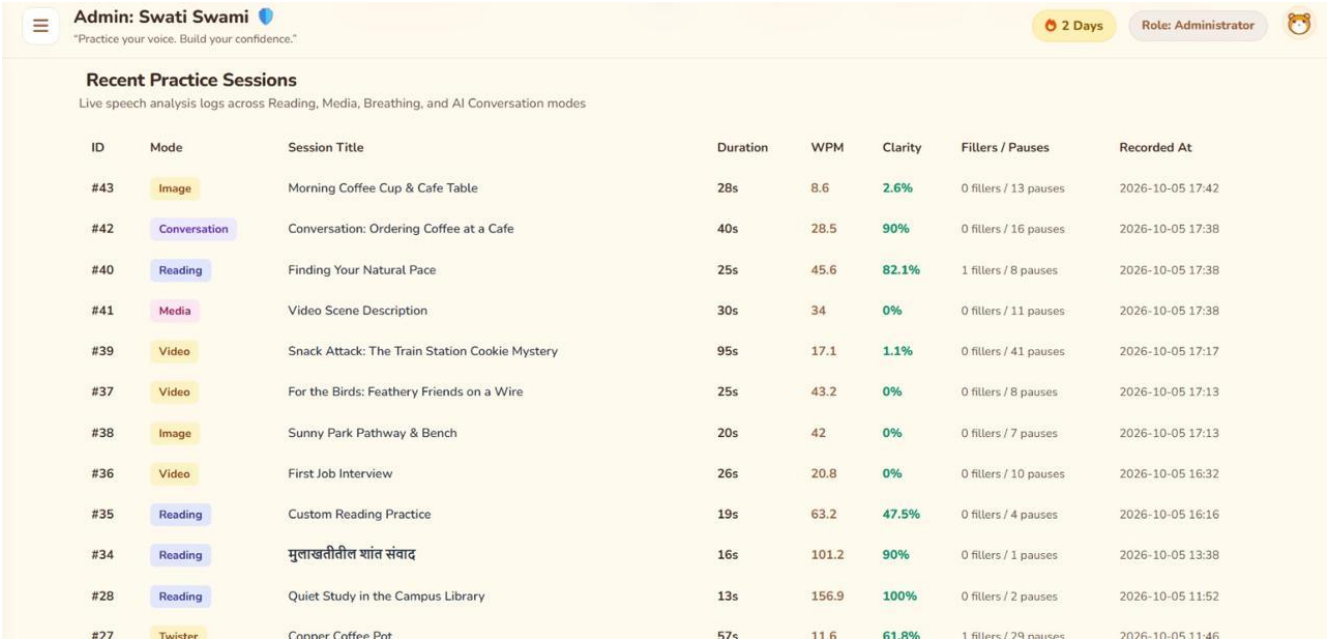
Login Page:-



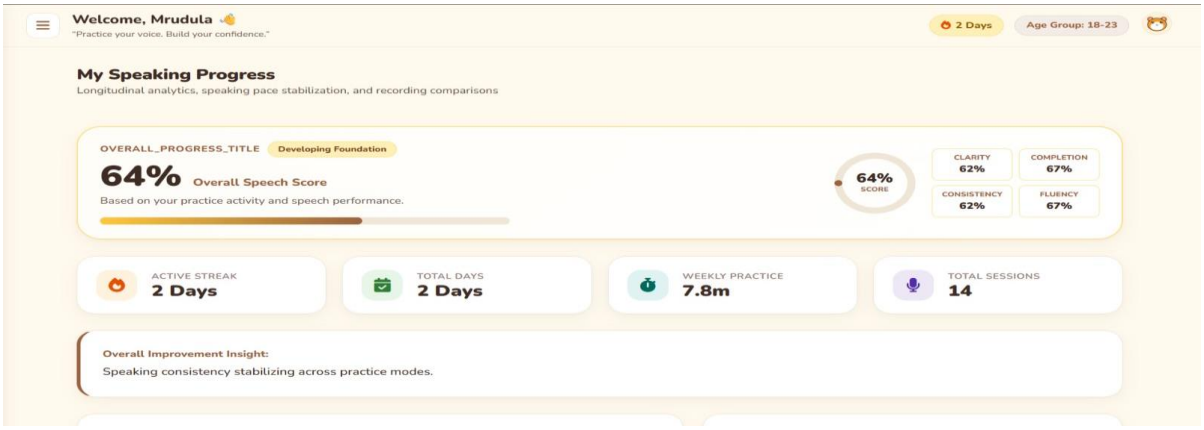
DashBoard:-



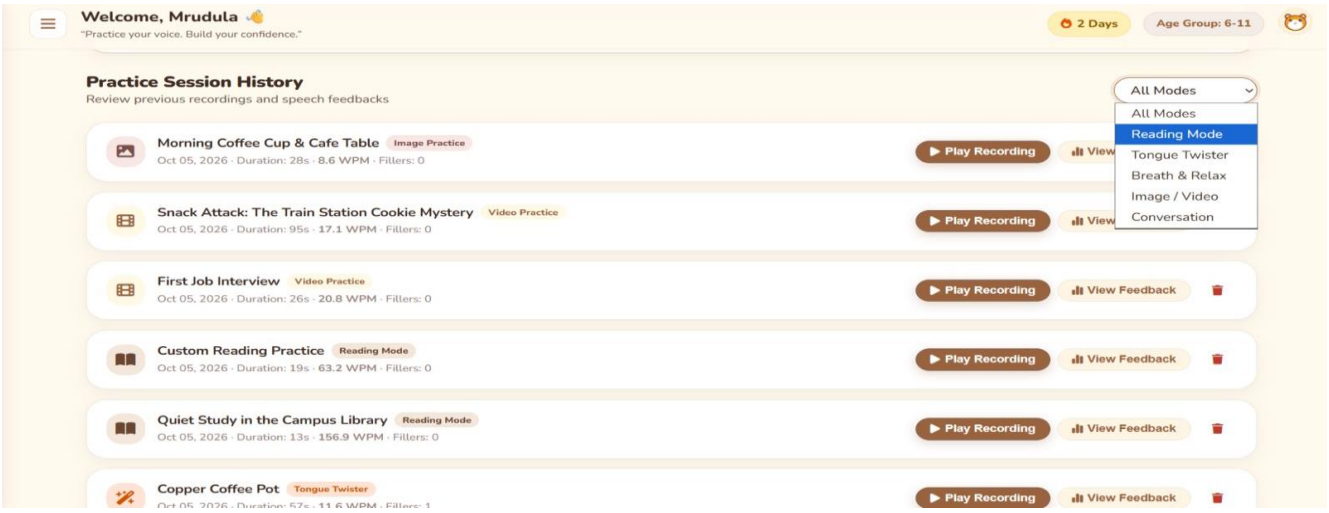
Admin Dashboard :-



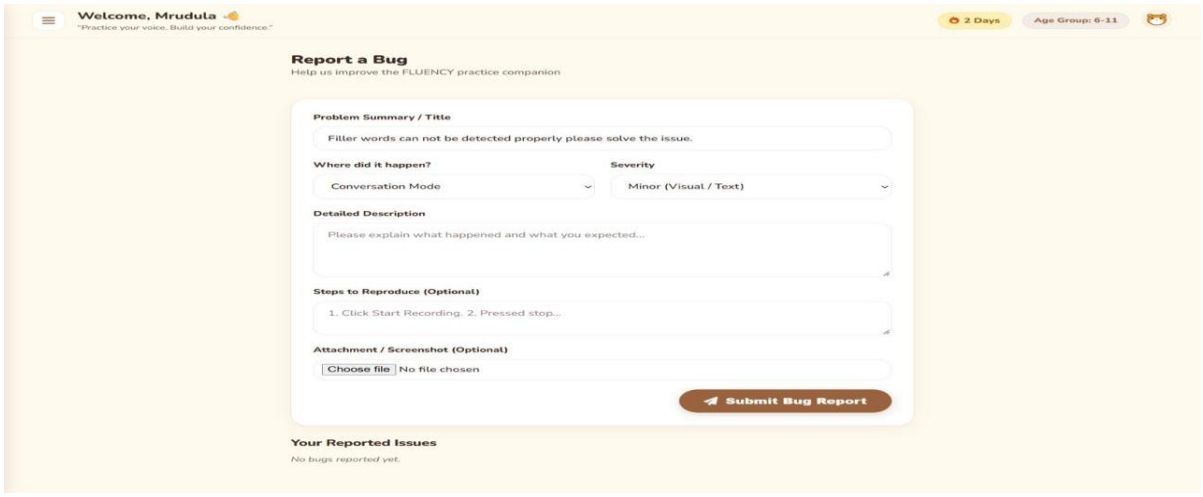
My Progress:-



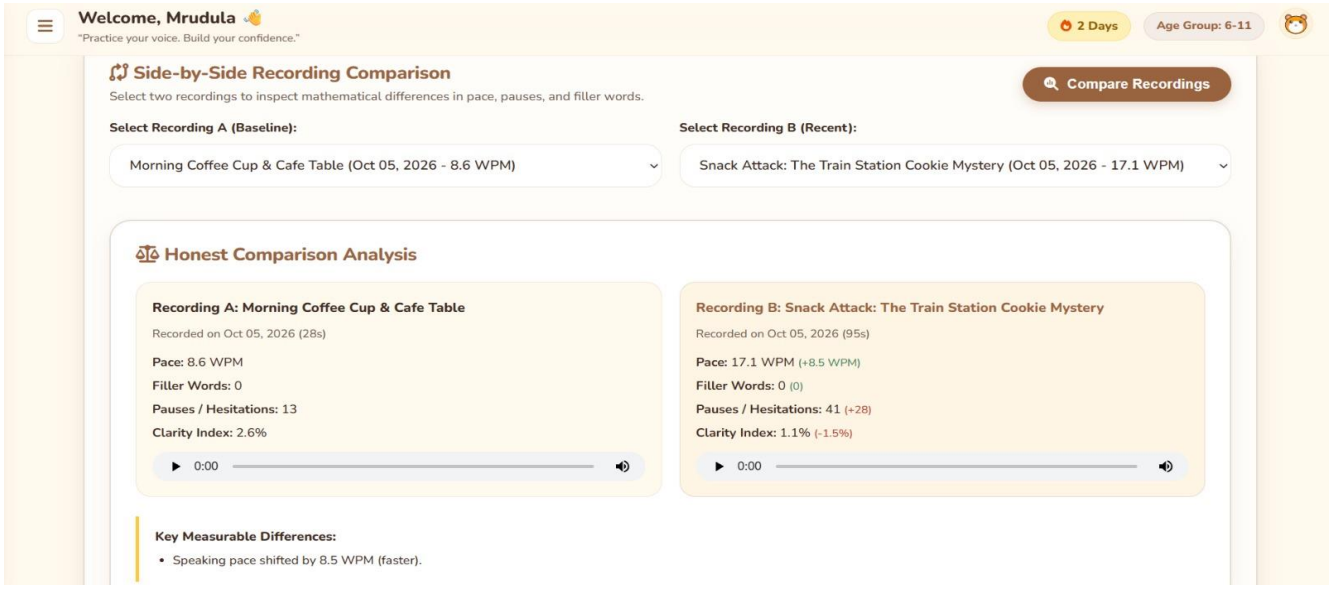
Practice Session History:-



Report and Bug:-



Compare Audio:-



Coding and Implementation:-

templates > index.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>FLUENCY — Practice your voice. Build your confidence.</title>

    <!-- Google Fonts: Nunito -->
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Nunito:ital,wght@0,300;0,400;0,500;0,600;0,700;0,800;0,900;1,400;1,700&dis

    <!-- FontAwesome 6 Icons -->
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.1/css/all.min.css">

    <!-- Chart.js for Progress Analytics -->
    <script src="https://cdn.jsdelivr.net/npm/chart.js@4.4.1/dist/chart.umd.min.js"></script>

    <!-- Application Stylesheets -->
    <link rel="stylesheet" href="/static/css/main.css">
    <link rel="stylesheet" href="/static/css/modes.css">
</head>
<body>

<div id="app-container">

    <!-- ================================================================
         SIDEBAR NAVIGATION (DEFAULT CLOSED, SLIDES IN/OUT VIA MENU BUTTON)
         ================================================================ -->
    <aside class="fluency-sidebar" id="fluency-sidebar">
        <div class="sidebar-brand">
            <div class="brand-mascot-badge" id="sidebar-mascot-avatar" data-route="dashboard">
                <!-- Mascot Injected via JS -->
            </div>
            <div class="brand-text">
                <h2>FLUENCY</h2>
```

app.py > ...

```python
import os
import sys
import json
import datetime
from dotenv import load_dotenv
from flask import Flask, request, jsonify, render_template, session, send_from_directory
from flask_cors import CORS
from models import db, User, PracticeSession, ConversationSession, UserProgressSnapshot, BugReport
from speech_analyzer import analyze_speech_data
from conversation_engine import generate_conversation_reply
import content_data

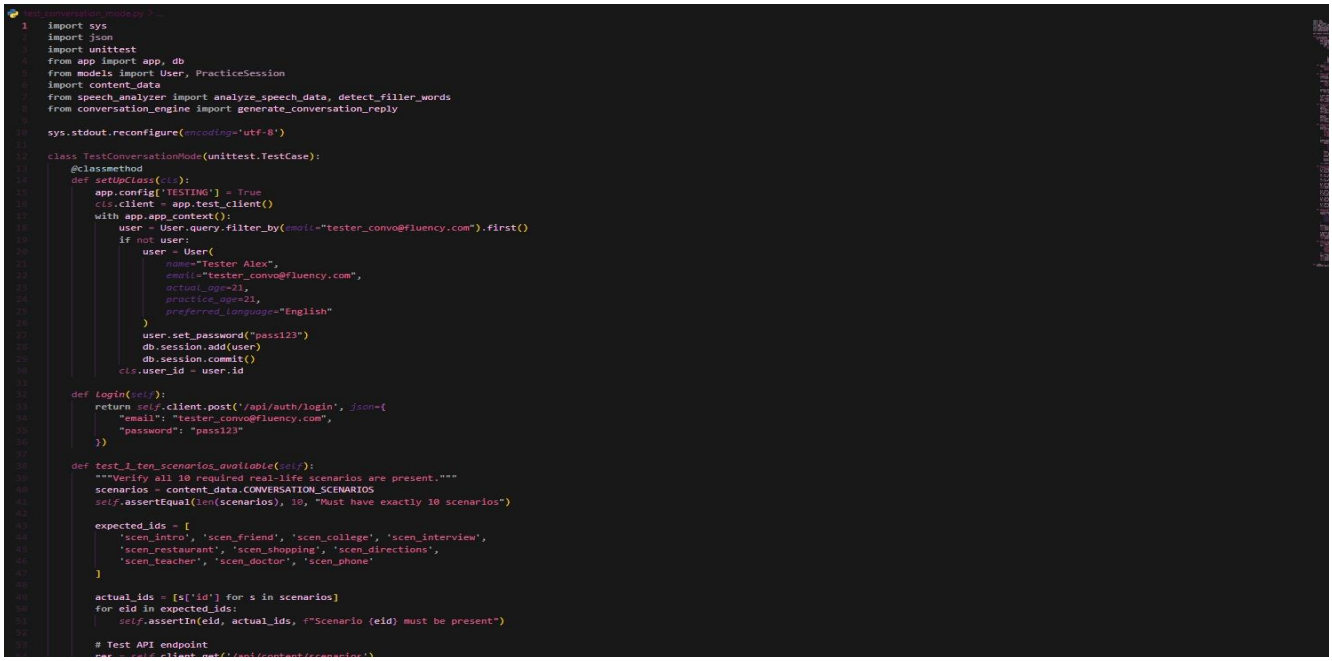
# Load environment variables from .env file
load_dotenv()

app = Flask(__name__, static_folder="static", template_folder="templates")
app.secret_key = os.environ.get("SECRET_KEY", os.environ.get("FLASK_SECRET_KEY", "fluency-secure-random-production-key-2026-speec

# Database Configuration from Environment Variables (MySQL 8.0)
DB_TYPE = os.environ.get("DB_TYPE", "mysql").lower()
DB_USER = os.environ.get("DB_USER", "root")
DB_PASSWORD = os.environ.get("DB_PASSWORD", "Kulkarni123")
DB_HOST = os.environ.get("DB_HOST", "127.0.0.1")
DB_PORT = os.environ.get("DB_PORT", "3306")
DB_NAME = os.environ.get("DB_NAME", "fluency_db")

# Fixed Admin Authentication Credentials from Environment Variables
ADMIN_ID = os.environ.get("ADMIN_ID", "MruSwatiSwami")
ADMIN_PASSWORD = os.environ.get("ADMIN_PASSWORD", "mruKulkarni1")

if DB_TYPE == "mysql":
    app.config['SQLALCHEMY_DATABASE_URI'] = f"mysql+pymysql://{DB_USER}:{DB_PASSWORD}@{DB_HOST}:{DB_PORT}/{DB_NAME}?charset=utf8m
else:
    app.config['SQLALCHEMY_DATABASE_URI'] = 'sqlite:///database.db'
```



```python
import sys
import json
import unittest
from app import app, db
from models import User, PracticeSession
import content_data
from speech_analyzer import analyze_speech_data, detect_filler_words
from conversation_engine import generate_conversation_reply

sys.stdout.reconfigure(encoding='utf-8')

class TestConversationMode(unittest.TestCase):
    @classmethod
    def setUpClass(cls):
        app.config['TESTING'] = True
        cls.client = app.test_client()
        with app.app_context():
            user = User.query.filter_by(email="tester_convo@fluency.com").first()
            if not user:
                user = User(
                    name="Tester Alex",
                    email="tester_convo@fluency.com",
                    actual_age=21,
                    practice_age=21,
                    preferred_language="English"
                )
                user.set_password("pass123")
                db.session.add(user)
                db.session.commit()
            cls.user_id = user.id

    def login(self):
        return self.client.post('/api/auth/login', json={
            "email": "tester_convo@fluency.com",
            "password": "pass123"
        })

    def test_1_ten_scenarios_available(self):
        """Verify all 10 required real-life scenarios are present."""
        scenarios = content_data.CONVERSATION_SCENARIOS
        self.assertEqual(len(scenarios), 10, "Must have exactly 10 scenarios")

        expected_ids = [
            'scen_intro', 'scen_friend', 'scen_college', 'scen_interview',
            'scen_restaurant', 'scen_shopping', 'scen_directions',
            'scen_teacher', 'scen_doctor', 'scen_phone'
        ]

        actual_ids = [s['id'] for s in scenarios]
        for eid in expected_ids:
            self.assertIn(eid, actual_ids, f"Scenario {eid} must be present")

        # Test API endpoint
        res = self.client.get('/api/content/scenarios')
```



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import json
import unittest
from app import app, db
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import content_data
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                    practice_age=21,
                    preferred_language="English"
                )
                user.set_password("pass123")
                db.session.add(user)
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```

References:-

1. Mozilla Developer Network (MDN). **HTML, CSS and JavaScript Documentation.** Available at: https://developer.mozilla.org/
2. Python Software Foundation. **Python Documentation.** Available at: https://docs.python.org/
3. Pallets Projects. **Flask Documentation.** Available at: https://flask.palletsprojects.com/
4. Oracle. **MySQL 8.0 Reference Manual.** Available at: https://dev.mysql.com/doc/
5. W3C. **Web Standards and Web Technologies Documentation.** Available at: https://www.w3.org/
6. PlantUML. **PlantUML Documentation – UML Diagrams.** Available at: https://plantuml.com/
7. MySQL. **MySQL Workbench Documentation.** Available at: https://dev.mysql.com/doc/workbench/en/
8. Python Package Index (PyPI). **Python Packages and Libraries Documentation.** Available at: https://pypi.org/
9. Web Speech API. **MDN Web Speech API Documentation.** Available at: https://developer.mozilla.org/en-US/docs/Web/API/Web_Speech_API
10. **FLUENCY Project Source Code and Database Implementation.** Developed as part of the academic project using HTML, CSS, JavaScript, Python/Flask and MySQL.

"Practice Your voice .Build Your Confidence."



Presented By:-

Mrudula Kulkarni

Swati Patil

Swami Deore

TY B.Sc.Computer Science (A)

Ashoka Center For Business and Computer Studies,Nashik